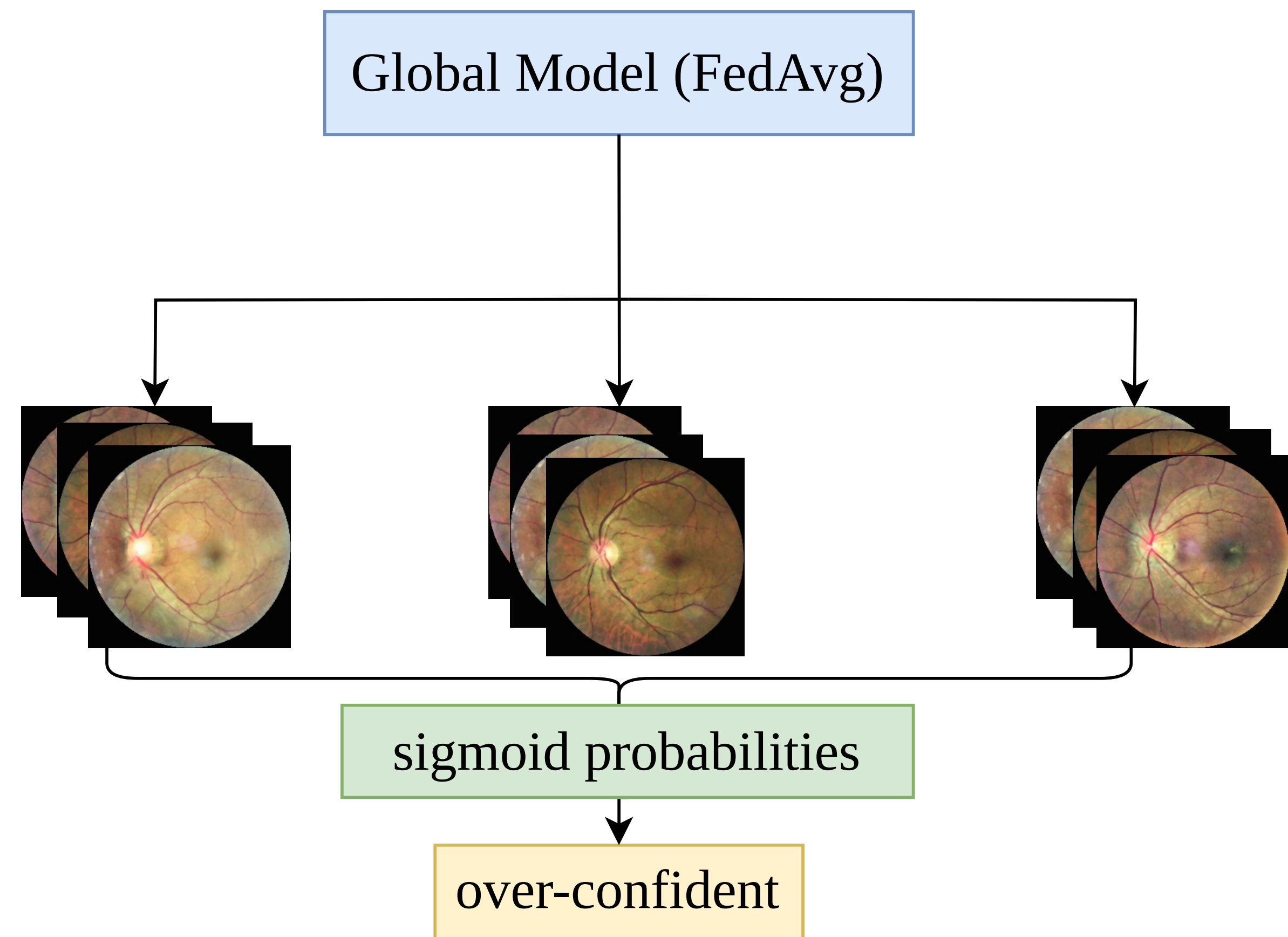
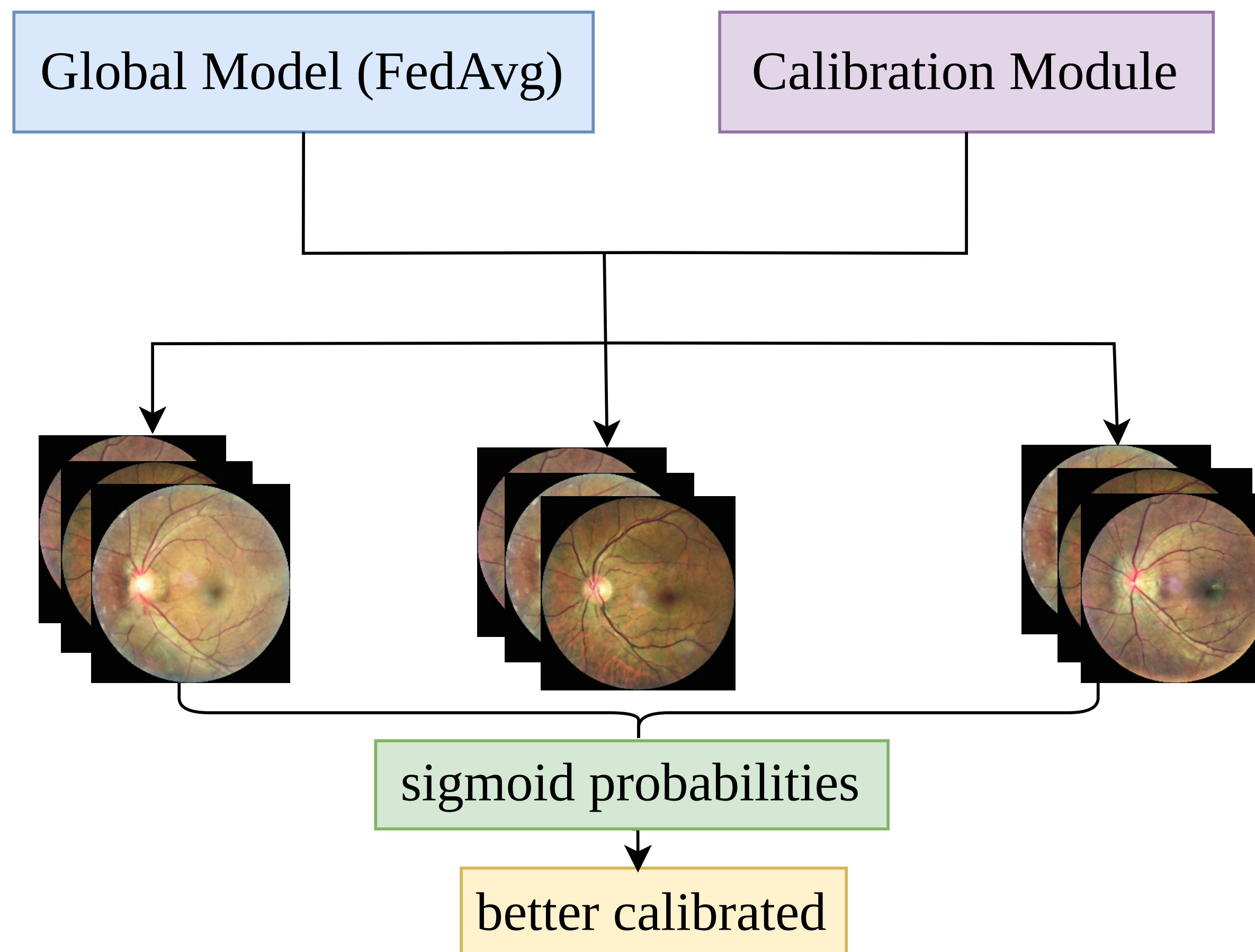


(a) Conventional FedAvg



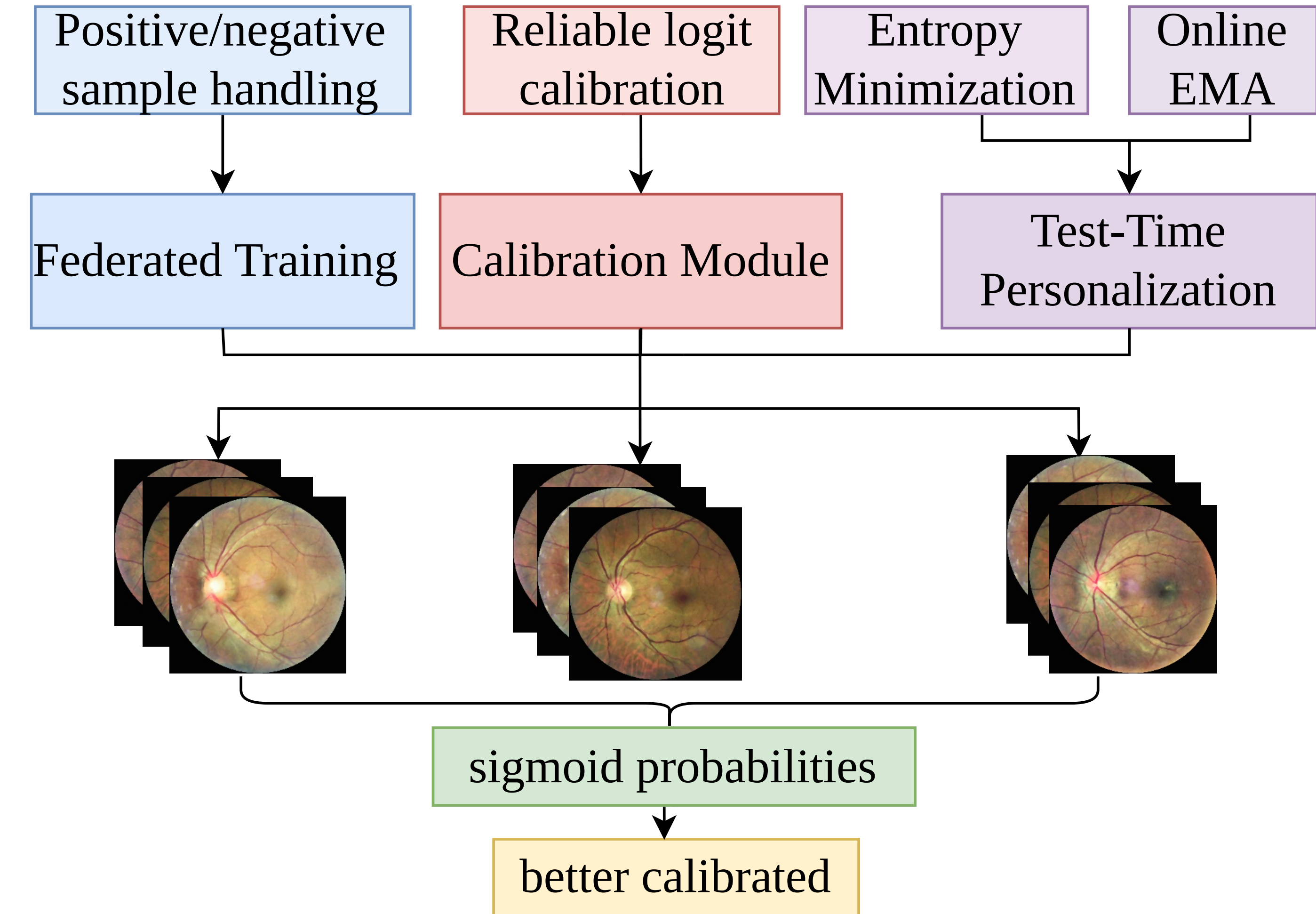
- ✗ Miscalibrated under domain shift
- ✗ No adaptation / long-tail bias

(b) FedAvg + Calibration (TS)



- ✗ No adaptation for new domain
- ✗ Still long-tail bias

(c) Ours (RAL + BVS + TTP)



- ✓ Long-tail aware training
- ✓ Reliable calibration
- ✓ Online test-time adaptation
- ✓ Personalized prediction under shift